

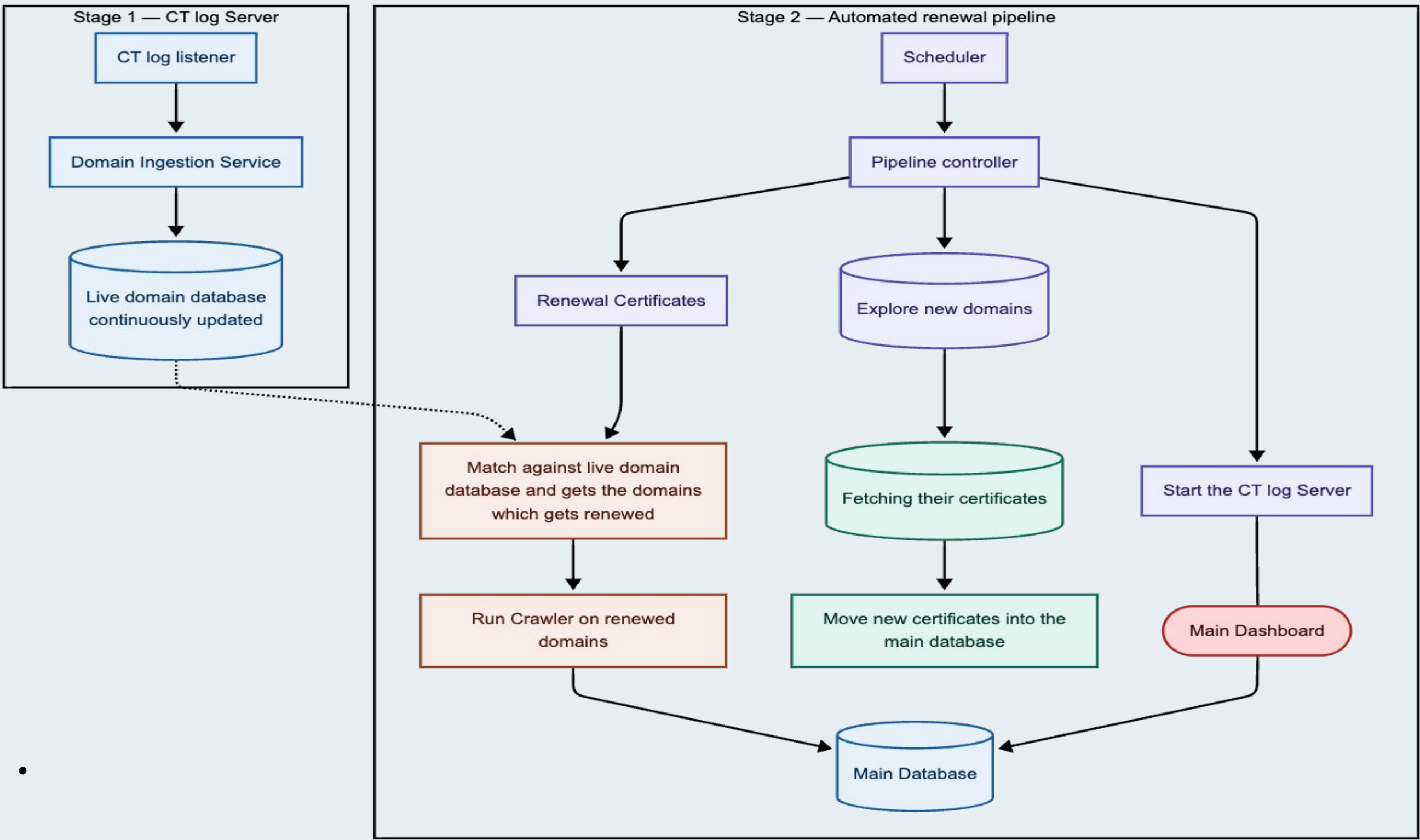


SSL Certificates Health

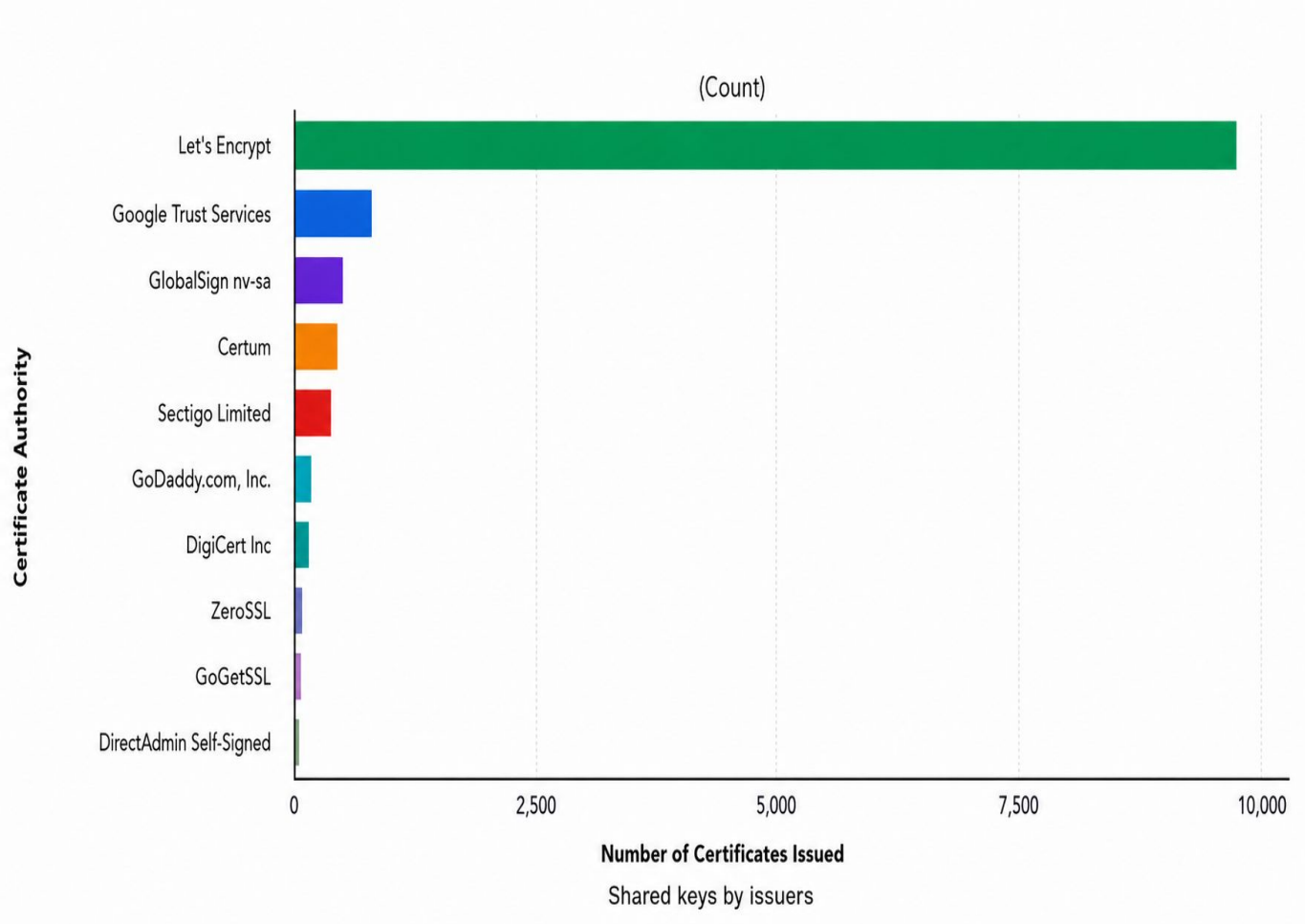
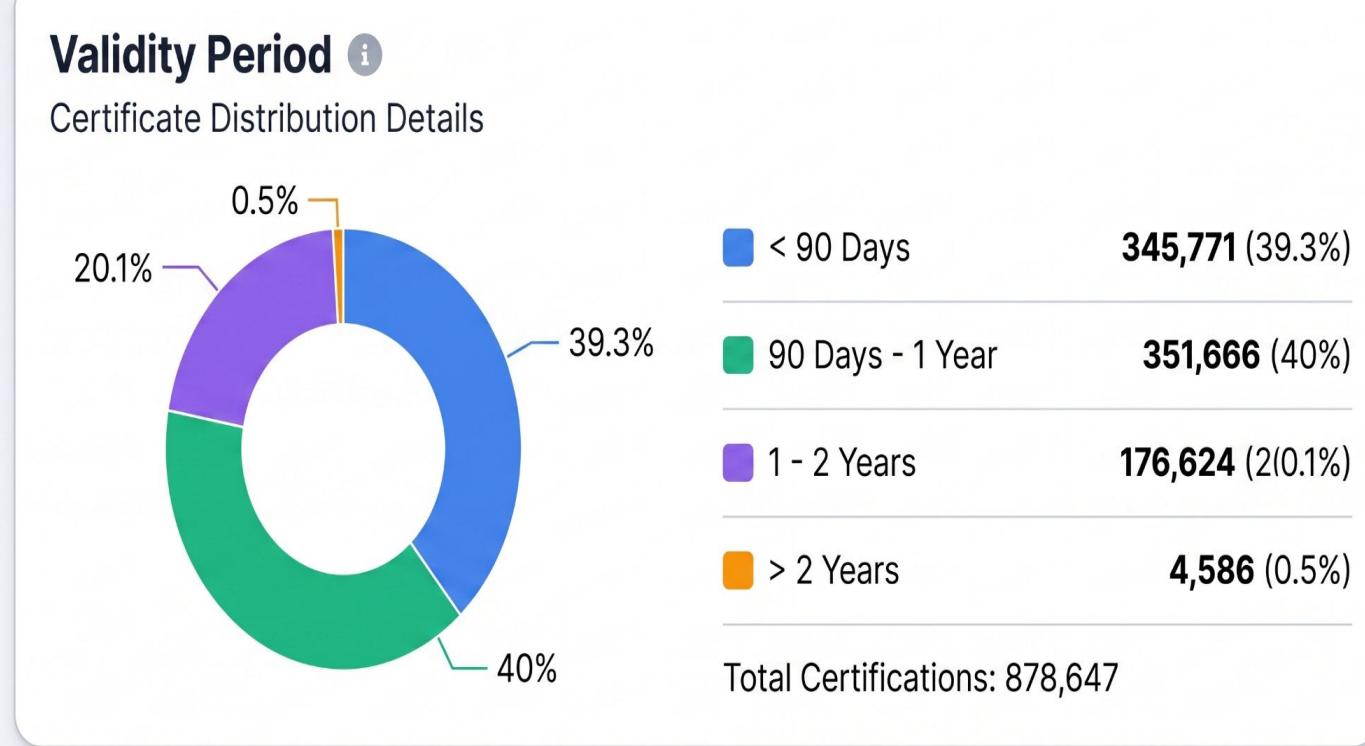
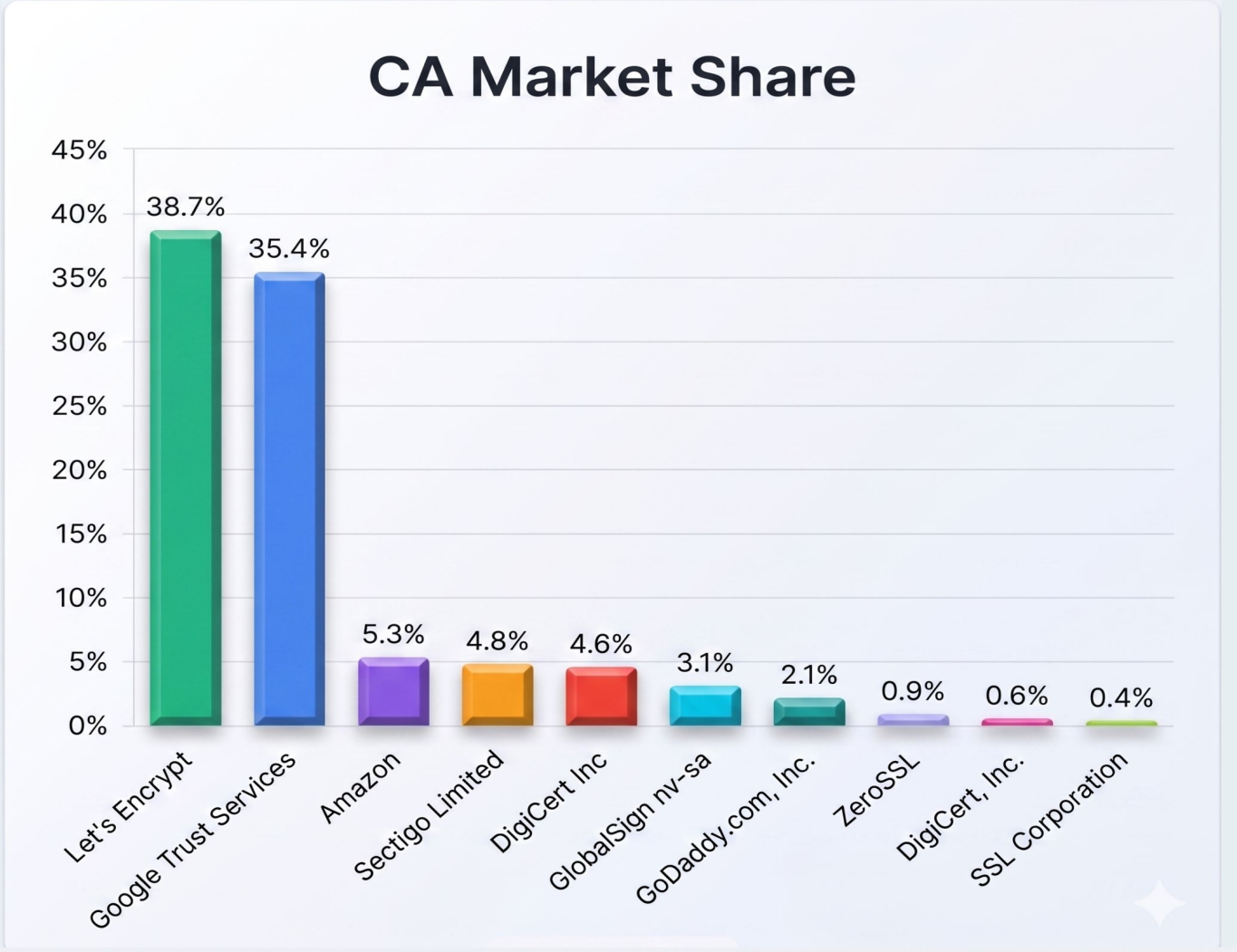
1 Introduction

TLS certificates are used to ensure encrypted traffic over the internet. Some certificates use weak algorithms, some are expired, and others reuse the same public keys across multiple websites. This makes websites vulnerable to attacks. There is **no simple tool** that analyzes SSL certificates fields, ranks CAs and TLDs based on market share, and highlights risky certificates in a clear and accessible way. Our project solves this with a fully automated system that continuously updates and expands our dataset using **CT logs**. All findings are displayed on a near real-time dashboard for clear, up-to-date visibility.

2 Methodology



3 Findings



Signature Strength by Issuer				
Issuer	RSA-2048	RSA-4096	ECDSA-256	ECDSA-384
Let's Encrypt	215.2k	37.9k	71.2k	10.5k
Google Trust Services			303.3k	
Amazon	46.1k			
Sectigo Limited	36.2k			
DigiCert Inc	33.0k			
GlobalSign nv-sa	20.8k			
GoDaddy.com, Inc.	18.0k			

- Let's Encrypt dominates the certificate authority market, holding 38.7% share, followed closely by Google Trust Services (35.4%), while all other CAs individually account for less than 6%.
- Certificate validity periods are predominantly short-term, with 79.3% of certificates valid for one year or less, reflecting industry adoption of shorter certificate lifecycles.
- RSA-2048 remains the most widely used signature algorithm, particularly among major issuers such as Let's Encrypt, Amazon, Sectigo, and DigiCert.
- Google Trust Services stands out for its extensive use of ECDSA-256 signatures, issuing approximately 303.3k certificates, indicating a stronger shift toward modern cryptographic algorithms.

4 Tech Stack



5 Future Work

- Predictive ML Analytics:** Using machine learning to predict domain security risks based on historical certificate trends.
- Security-Based Domain Ranking:** Launching a custom benchmark index (similar to Tranco) that ranks the web's safest domains based on cryptographic strength.
- On-Demand Handshake Visualizer:** Allowing users to input any domain to generate an interactive, step-by-step mapping of the TLS/SSL handshake.

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